In simple words, DK counts how many ancestors there are for the node holding the k-th smallest key, NK; yet another way to see this number is: how many nodes there are in the path from the tree's root to NK?

Formally:

$$D_k = \sum_{i=0}^{n-1} N_i \text{ is an ancestor of } N_k$$

We can denote the even Ni is an ancestor of NK with the indicator (a binary variable) AKi, and then the expected value for DK becomes:

$$E\left[D_k\right] = \sum_{i=0}^{n-1} 1 \cdot P\left(N_i \text{ is an ancestor of } N_k\right) = \sum_{i=0}^{n-1} P\left(A_k^i\right)$$

For our randomized treap, we will define a random variable DK for the depth of a given node, NK: in simple words, DK counts how many ancestors there are for the node holding the k-th smallest key in the tree.